Name: ____

A Level Mathematics : Pure Mathematics

Topic 8 :Integration -Integration as the limit of a sum

Date:

Time:

Total marks available:

Total marks achieved: ______

© 2025 Exam Papers Practice. All Rights Reserved

Questions

Q1.

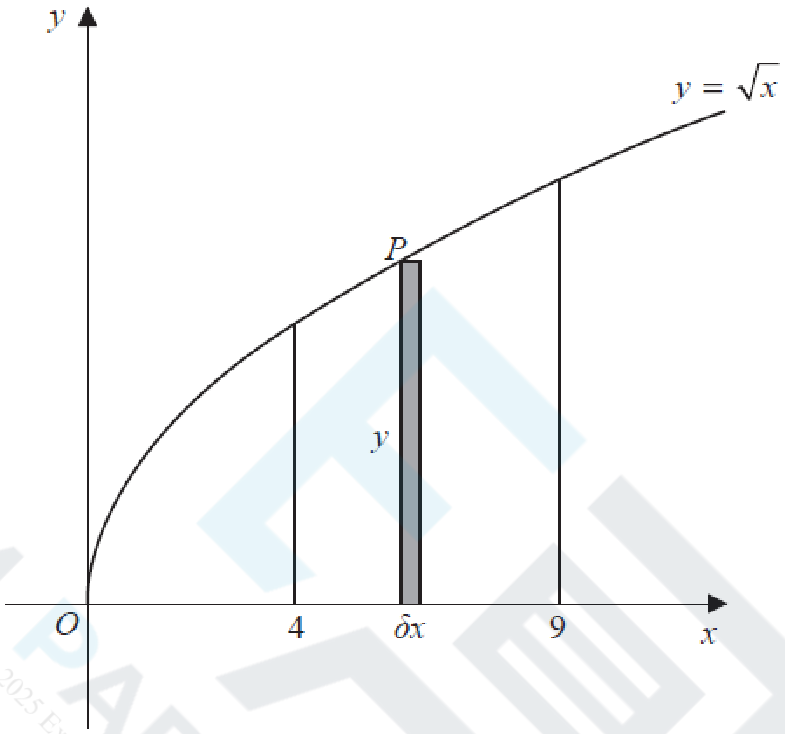

Figure 3

Figure 3 shows a sketch of the curve with equation $y = \sqrt{x}$

The point P(x, y) lies on the curve.

The rectangle, shown shaded on Figure 3, has height y and width δx.

Calculate

$$\lim_{\delta x \to 0} \sum_{x=4}^{9} \sqrt{x}\,\delta x$$

(Total for question = 3 marks)

© 2025 Exam Papers Practice. All Rights Reserved

EXAM PAPERS PRACTICE

Q2.

(a) Given that

$$\frac{x^2 + 8x - 3}{x + 2} \equiv Ax + B + \frac{C}{x + 2} \qquad x \in \mathbb{R}\ x \neq -2$$

find the values of the constants *A*, *B* and *C*

(3)

(b) Hence, using algebraic integration, find the exact value of

$$\int_0^6 \frac{x^2 + 8x - 3}{x + 2}\,\mathrm{d}x$$

giving your answer in the form *a* + *b* ln 2 where *a* and *b* are integers to be found.

(4)

(Total for question = 7 marks)

Q3.

The table below shows corresponding values of *x* and *y* for $y = \sqrt{\frac{x}{1 + x}}$

The values of *y* are given to 4 significant figures.

x	0.5	1	1.5	2	2.5
y	0.5774	0.7071	0.7746	0.8165	0.8452

(a) Use the trapezium rule, with all the values of y in the table, to find an estimate for

$$\int_{0.5}^{2.5} \sqrt{\frac{x}{1 + x}}\,\mathrm{d}x$$

giving your answer to 3 significant figures.

(3)

(b) Using your answer to part (a), deduce an estimate for $\int_{0.5}^{2.5} \sqrt{\frac{9x}{1 + x}}\,\mathrm{d}x$

(1)

Given that

$$\int_{0.5}^{2.5} \sqrt{\frac{9x}{1 + x}}\,\mathrm{d}x = 4.535 \text{ to 4 significant figures}$$

© 2025 Exam Papers Practice. All Rights Reserved

(c) comment on the accuracy of your answer to part (b).

(1)

(Total for question = 5 marks)

Q4.

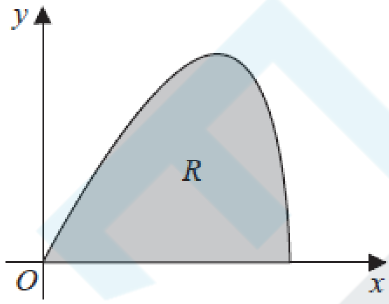

Figure 3

The curve shown in Figure 3 has parametric equations

$$x = 6\sin t \qquad y = 5\sin 2t \qquad 0 \leqslant t \leqslant \frac{\pi}{2}$$

The region *R*, shown shaded in Figure 3, is bounded by the curve and the *x*-axis.

(a) (i) Show that the area of *R* is given by $\int_0^{\frac{\pi}{2}} 60\sin t\cos^2 t \, dt$

(3)

(ii) Hence show, by algebraic integration, that the area of *R* is exactly 20

(3)

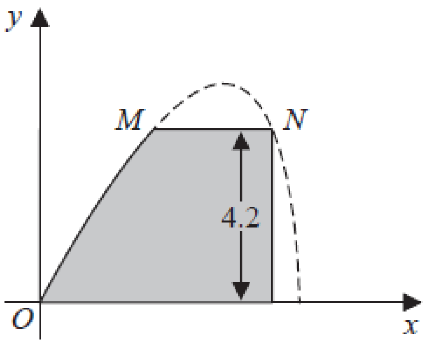

Figure 4

Part of the curve is used to model the profile of a small dam, shown shaded in Figure 4.
Using the model and given that

- x and y are in metres
- the vertical wall of the dam is 4.2 metres high
- there is a horizontal walkway of width MN along the top of the dam

(b) calculate the width of the walkway.

(5)

(Total for question = 11 marks)

© 2025 Exam Papers Practice. All Rights Reserved